

Goldman Sachs India Hackathon

2026 | QUANTITATIVE TRACK

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ETF Proxy Models

01 The problem

$$ETF_{i,t} = \sum_j w_j Proxy_{j,t}$$

- Proxy instruments include Proxy ETFs, Bonds, Futures, CDX.
- We ignore Proxy ETFs for risk estimation

NAV weights

Risk weights

Hedge

Linear Regression?

02 NAV and Risk Estimation

Overfitting on noisy returns
→ **Ridge regression**

Tiny weights → transaction costs may exceed profit
→ **Lasso (sparse weights)**

Unclean data — extreme outliers
→ **Winsorization**

Another step to ensure outliers do not distort the model
→ **Huber regressor**

Avoid Ridge's tiny-weight problem again
→ **Elastic Net as selector**

03 The Hedge

$$\text{Huber Loss : } L_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } |a| \leq \delta \\ \delta\left(|a| - \frac{1}{2}\delta\right) & \text{for } |a| > \delta \end{cases}$$

Elastic-Net selector + Huber Regressor

Fit on target $y = -\text{Portfolio PnL}$
→ **chosen to maximize HER**

- Two ways to fit – with and without intercept. The fundamental difference is a drifting portfolio .

Improvements

The Hedge -

- No transaction cost penalty — incorporate into the objective function of an optimizer or scale data by $1/(\text{transaction_cost})$.
- The instrument selector is blind to transaction costs. Dropping it and incorporating transaction cost into an optimizer would be better.

NAV and Risk Model -

- Account for the DV01 metric – longer duration bonds show more DV01 than shorter durations.
- Regime detection – improve upon the plan for high volatility regimes

Regime Navigator

Bermudan Swaption Pricing

01 The problem

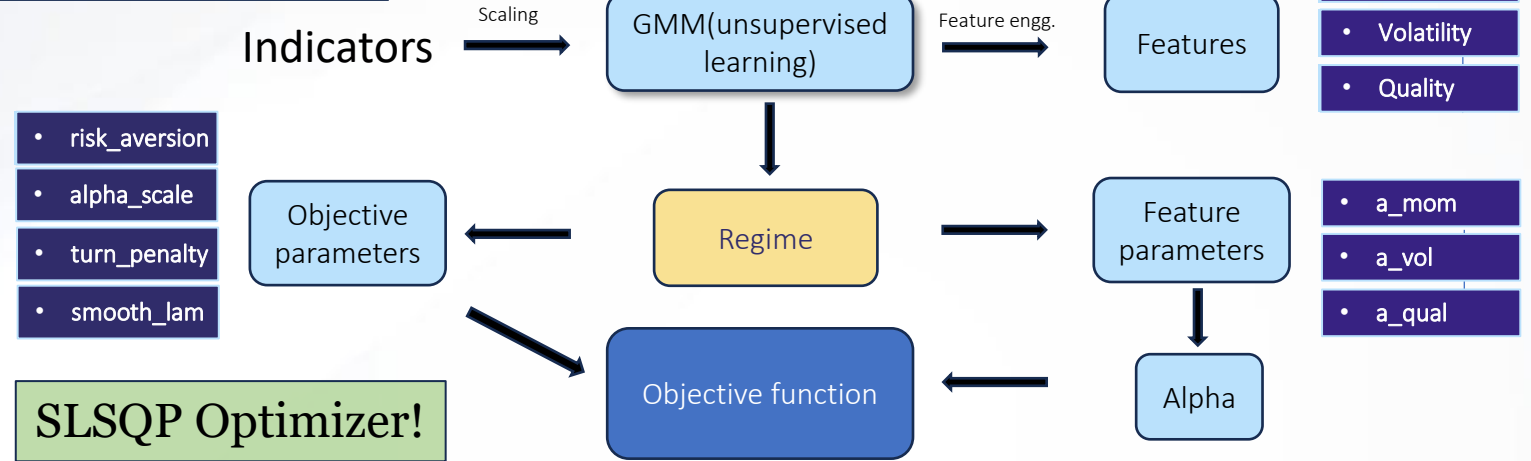
Given 100 securities –

- Build a regime prediction framework
- Build a regime-adaptive portfolio

Some constraints -

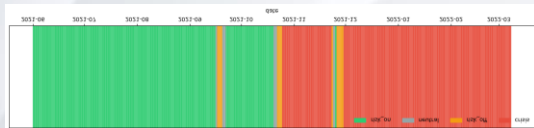
- Net exposure : $0.85 \leq \sum w_i \leq 1.1$
- Max single position : $|w_i| \leq 0.1$
- Short exposure : ≥ -0.3

02 The architecture

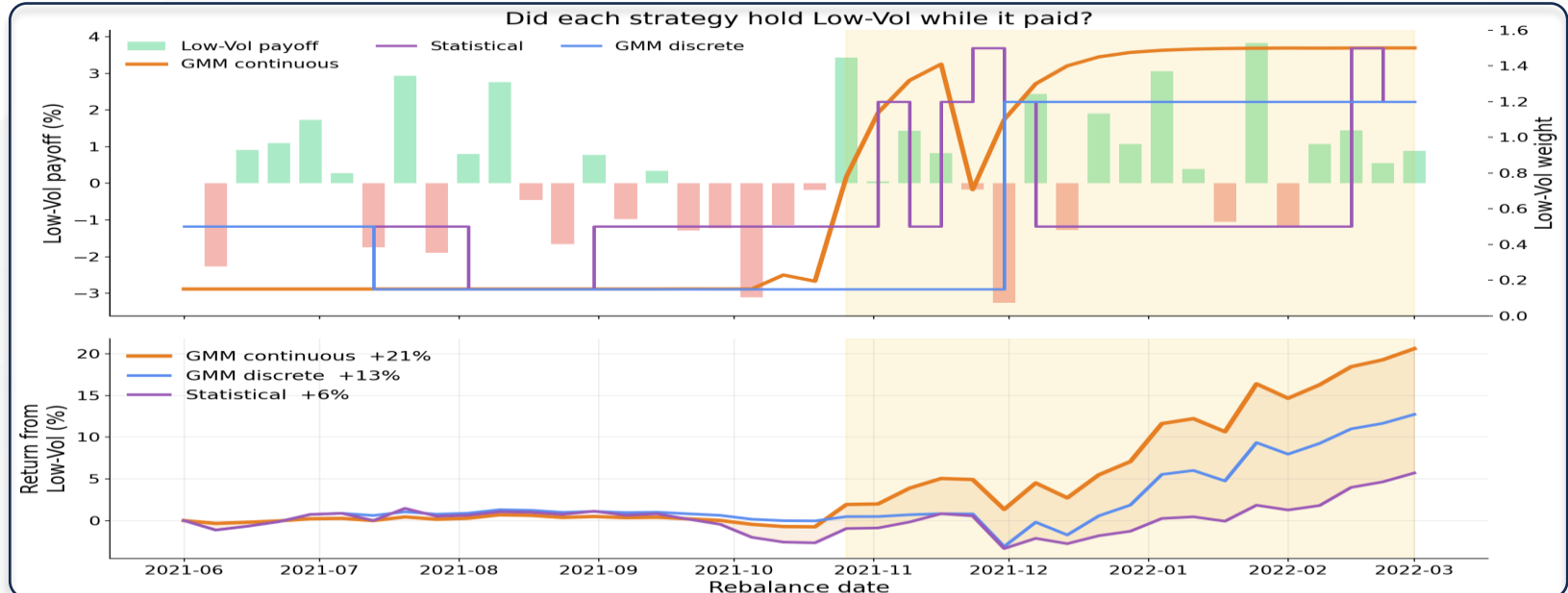


03 Continuous Regime Prediction

- Find probability of belonging to stress regime by using GMM.
- Interpolate all parameters based on a continuous regime spectrum.

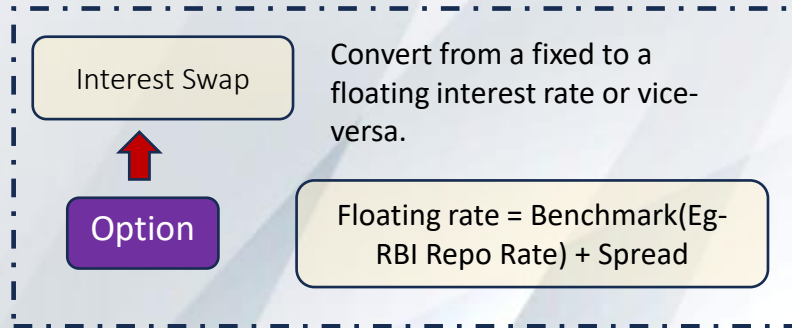


Regime Whipsawing!



Bermudan Swaption Pricing

01 The problem



- European – Single exercise date
- American – Continuous exercise period
- Bermudan – Discrete exercise dates

Payer Swaption – Reserves the right to pay fixed rate and receive floating rate

Yield Curve

Discount Factor

$$P(0, T) = \exp(-r(T) \cdot T)$$

Forward Rate

$$f(0, T_i) = -\frac{\ln P(0, T_{i+1}) - \ln P(0, T_i)}{T_{i+1} - T_i}$$

02 The Hull-White model

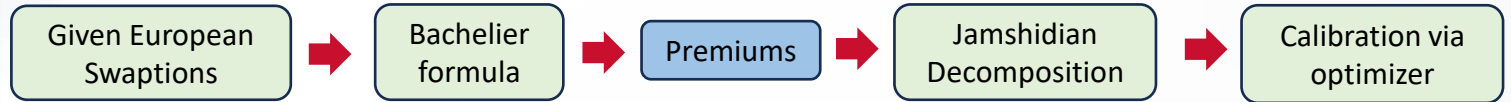
$$dr_t = (\theta(t) - ar_t)dt + \sigma dW_t$$

a – reversion constant

σ – volatility

$\theta(t)$ – drift

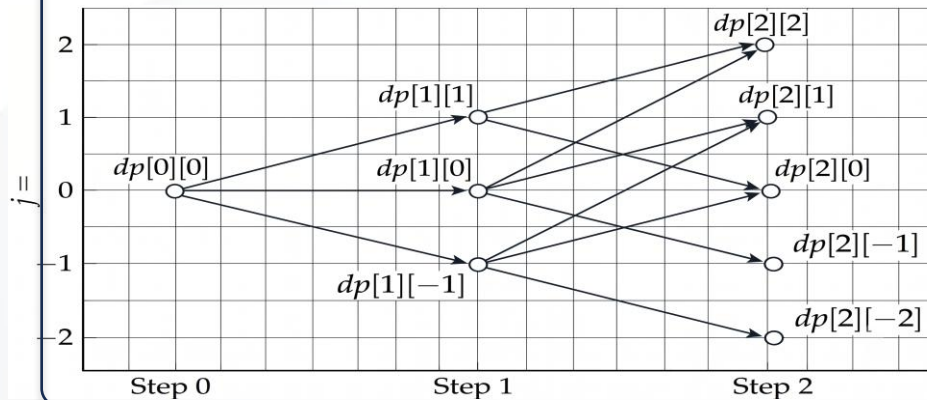
$dW_t \sim N(0, dt)$



03 Pricing the Bermudan Swaption

$$dp_{\text{cont}}[i][j] = e^{-r_{i,j}\Delta t} (p_u \cdot dp[i+1][j+1] + p_m \cdot dp[i+1][j] + p_d \cdot dp[i+1][j-1])$$

$$dp[i][j] = \max(IV_{i,j}, dp_{\text{cont}}[i][j])$$



Trinomial Tree

- $IV_{i,j}$ is computed by Hull-White
- Transition probabilities are computed by using the reversion constant.
- Increase number of transitions. (pentanomial)

04 Vega/Delta

Vega – Bump each vol by +1 bp , recalibrate, reprice and compute the difference

Delta – Bump each zero rate by +1 bp , recalibrate, reprice and compute the difference